

Shrihari Gokulachandran

Austin, TX, USA | shariethernet@outlook.com | [LinkedIn](#) | [GitHub](#)

Profile

Digital Design Engineer with experience designing AI/ML accelerator hardware, memory controllers and low-power digital IPs. At Silicon Labs, served as the sole architect and RTL designer for an early-stage edge ML inference accelerator IP, performing modeling and profiling studies, defining and implementing microarchitecture in RTL. Experienced in SystemVerilog RTL design, architectural modeling, workload-driven design exploration, and hardware/software co-design. Additional experience includes research on workload-driven power-management and synthesis optimization for heterogeneous systems, along with RTL-to-GDS implementation and post-silicon bring-up of a TSMC 65nm compute-near-memory MPW test chip.

Skills

RTL DESIGN: SystemVerilog, Verilog, TL-Verilog

PROGRAMMING: Python, C, Bash, TCL

EDA TOOLS: Synopsys: VCS, Verdi, PrimeTime/PrimePower (PTPX), Design Compiler; Cadence: Xcelium, Verisium, Genus, Innovus, JasperGold, Conformal LEC; Siemens: QuestaSim, Formal-Lint-CDC, RDC; Vivado; MATLAB

Education

THE UNIVERSITY OF TEXAS AT AUSTIN, Texas, USA

Ph.D. - Architecture, Computer Systems, and Embedded Systems (ACSES) track, GPA 3.9/4 August 2022 – Present

- Workload-aware power/energy optimization for FD-SOI-based heterogeneous edge computing platforms.

MS - Electrical and Computer Engineering, GPA 3.9/4 August 2022 – May 2024

- Courses: Computer Architecture, SoC Design, ASIC Design, Verification of Digital Systems, High Speed Computer Arithmetic, Hardware Architectures for Machine Learning, Embedded IoT

PSG COLLEGE OF TECHNOLOGY, Coimbatore, India August 2018 – May 2022

Bachelor of Engineering - Electronics and Communication Engineering, GPA: 9.92/10

- Summa Cum Laude - Class of 2022, Best Outgoing Student and Best Bachelor Thesis Award
- Related coursework: Computer Architecture, Digital Design, VLSI Design, Embedded Systems, Digital Signal Processing, Advanced Physical Design using 130nm PDK (VSD), Building a RISC-V CPU (Linux Foundation)

Experience

DIGITAL DESIGN ENGINEER | SILICON LABS July 2024 - Present

- Sole architect and RTL designer for an early-stage edge ML inference accelerator IP, from workload profiling and NPU benchmarking to microarchitecture definition and RTL implementation.
- Defined a memory-aware accelerator architecture to maximize MAC utilization, reduce memory bandwidth, and improve throughput/latency across edge ML workloads.
- Architected and implemented features including on-the-fly convolution lowering, optimized buffering/tiling, back-to-back convolution dataflow, and affine re-quantization, achieving ~3× higher average MAC utilization versus evaluated commercial NPU baselines.
- Designed memory controller IP, and owned AHB/AXI/APB based register interface and bus level security generators for chip infrastructure and performed IP quality checks including lint, synthesis, LEC and trace-based power analysis.
- Received Silicon Labs’ AI award (Q2-2025) and Technical Symposium 2025 for novel architectural contributions in ML acceleration and development of internal agentic frameworks.

GRADUATE RESEARCH ASSISTANT | SLAM LAB - UNIVERSITY OF TEXAS AT AUSTIN Jan – May 2023, Sep 2023 – May 2024

- Developed workload-based power-management strategies (design-time, run-time DVFS vs DVBFS) for heterogeneous platforms (RISC-V based SoCs) on GlobalFoundries 22nm-FDSOI Technology.

RTL DESIGN ENGINEER - INTERN | SILICON LABS, Austin May 2023 – August 2023

- Developed an approximately timed architectural model of a low-power ML inference accelerator and profiled architectural variants and third-party accelerator baselines across MLPerf Tiny workloads.
- Contributed to early accelerator definition through workload-driven architectural studies, feature exploration, and initial microarchitecture development.

DESIGN VERIFICATION ENGINEER - INTERN | SILICON LABS, India

January 2022 – July 2022

- Developed golden-reference C models for AES modes used in QSPI interfaces and integrated them into a UVM based testbench, identifying critical IP level and security sub-system level bugs.
- Developed frameworks for migration and integration of legacy projects in Silicon Labs design environment.
- Worked on verification components for ARM Low Power Interface, Silicon Labs Peripheral Reflex System, AXI, SPI and formal verification bring-up for upcoming projects.

CPU DESIGN ENGINEER – INTERN | INCORE SEMICONDUCTORS, India

July 2021 – December 2021

- Architected and developed custom co-processor interconnects for RISC-V processors.
- Updated SoC-generation frameworks with interconnect generation support.

Research papers

- S. Gokulachandran et al., “Dynamic Voltage, Body Bias and Frequency Scaling for FD-SOI-Based Low-Power Edge Processors”, *Design Automation and Test in Europe (DATE)*, 2026.

Academic Projects

DARKNET ACCELERATOR: ZYNQ MPSOC BASED SYSTEM DESIGN

- Converted Darknet from FP32 to INT16, implemented multi-threading, developed a QEMU-SystemC TLM-based virtual platform prototype of a CPU-Accelerator system, developed Linux kernel modules, and hardware abstraction layers for hardware/software integration.
- Accelerated Darknet using a custom GEMM accelerator, and Xilinx DPU and deployed yolov3-tiny on Ultra96 FPGA.

COMPUTE NEAR MEMORY TEST CHIP – TSMC 65NM MPW TAPEOUT (RTL TO GDS + POST SILICON TESTING)

- Contributed to the end-to-end design cycle of a compute-near-memory test chip in TSMC 65 nm, spanning architecture exploration, RTL design, verification, and physical implementation for convolution, arithmetic, and logic operations.
- Performed post-silicon bring-up, and developed PCB, and FPGA-based emulation and test infrastructure.

LOGICNETS+: A FRAMEWORK FOR LOW LATENCY BINARY NEURAL NETWORK INFERENCE ON FPGAS

- Extended the LogicNets framework—originally designed for GEMM / fully connected layers—by enabling Conv2D, DepthwiseConv2D, and PointwiseConv2D evaluation using truth-table-based neuron implementations, achieving ~90% accuracy and <20 ns inference latency at 200 MHz on representative networks.

HARDWARE ACCELERATOR EXTENSION INTERFACE FOR RISC-V CPUS

- Researched open-source co-processor interfaces and designed a co-processor agnostic accelerator adapter for a RISC-V Core, based on OpenHW Group’s Core-V Extension Interface Specifications.

RPHAX - RAPID PROTOTYPING OF HARDWARE ACCELERATORS ON XILINX FPGAS (GOOGLE SUMMER OF CODE 2022)

- Designed and developed automation frameworks for IP packaging (AXI4-Lite, AXI Stream), block-design and bitstream generation enabling a single command, end-to-end Zynq based flow to prototype standalone IPs.

TRANSACTION LEVEL DESIGN OF MULTISTAGE RISC-V RV32IC CPU CORE AND PHYSICAL DESIGN ON 130NM PDK

- Designed a configurable 1–6-stage pipelined RISC-V CPU core generator using Transaction-Level Verilog and validated it with assembly programs. Completed RTL-to-GDSII implementation using open-source tools and the 130 nm SkyWater PDK, meeting timing and producing a DRC- and LVS-clean GDSII.

DESIGN VERIFICATION PROJECTS

- Formal verification of AXI-Stream and AXI-Stream-to-AXI-Lite bridges using Cadence JasperGold, including comparison of formal engines.
- Logic equivalence checking for the writeback stage of the ARM Amber core using Cadence Conformal.
- Sequential equivalence checking of a pipelined multiplier using Synopsys VC Formal and Verdi.
- Assertion-based verification of an Ethernet MAC module and Wishbone interface.
- UVM-based verification of a 32-bit ALU from ARM Amber Core.

AGENTIC RTL DEBUG FABRIC

- Built an MCP server with design and waveform introspection tools that enable a generalist coding agent to root-cause failing SystemVerilog simulations.
- Implemented EDA-tool-agnostic design-scope navigation, waveform inspection, driver/load causality tracing, value-aware X root-cause analysis, combinational-loop detection, and a signal-force/recompute layer for hypothesis testing by correlating elaborated RTL with waveform samples.